BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

LESSON 293 · Client-side versus server-side processing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an HTML form submit?

Q2. What does JavaScript textContent do?

Q3. What does GET generally do with form data?

Q4. A 2D array has 3 rows and 4 columns. Give total cells and last zero-based indexes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

client-side — Processing on user's browser/device.

server-side — Processing on service infrastructure.

validation — Checking data against rules.

authentication — Verifying user identity.

authorisation — Deciding permitted actions.

New learning

Understand — Client-side JavaScript can offer immediate feedback and responsive interfaces, but its validation can be bypassed or altered.

Apply — Server-side code enforces permissions, validates input against business rules and updates authoritative records or databases.

Evaluate — Processing choices trade off device load, server load, confidentiality, latency, accessibility and reliability; many systems use both.

Worked example

Why must a server check a submitted quantity even if the browser requires 1–5?

1. Predict first — An attacker can skip the browser form and send a request containing quantity=1000.

2. Explain — The server must enforce actual constraints; browser checks are useful but only improve convenience.

We Do · Practise with guidance

A pupil updates their profile photo. Decide which processing belongs in browser versus server.

```
Client: preview selection and format warning.
Server: verify account, check type/size, store safely.
```

Predict first

The browser can preview or resize an image to reduce traffic and improve usability.

Explain

The server verifies authorisation, validates the uploaded bytes and applies safe storage rules.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Compare client-side and server-side processing using form validation. [6]

Question 2

An online shop must deduct stock and charge for an order. Explain which operations cannot rely solely on a browser. [6]

Question 3

Evaluate a hybrid design for an accessible online examination practice system. [6]

Exit Ticket · Answer before checking

A server can trust form data solely because the browser used required.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Name/value fields to a server using a chosen HTTP method.

Q2. Changes text in a DOM node without parsing it as HTML.

Q3. Encodes it as query parameters in the request URL.

Q4. 12 cells; last index pair (2,3).

You Do pages · Independent questions

Q1. Browser offers fast feedback [2], can be bypassed [1]; server enforces authoritative rules [2], and may add network round-trip latency [1].

Q2. Authentication/authorisation [2], stock update [1], secure payment integration [1], transaction/integrity [1], server-side validation [1].

Q3. Client keyboard/visual feedback [1], client input checks [1], server stores progress [1], authenticates users [1], safeguards private results [1], trade-offs connectivity/load [1].

Exit Ticket · Model answer

False. Server validation is always necessary.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

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BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Do Now Recall previous learning | I Do Read new knowledge | We Do Practise with guidance | You Do Work independently | Review & Improve Check my understanding